The inductive reactance (X_L) varies directly as the frequency.

$$X_L = 2\pi f L \qquad (4.42)$$

where f is the frequency of the alternating current. For a steady current, $f = 0$. Therefore, $X_L = 0$. Thus an ideal inductor offers no resistance to steady DC current.

4.7.6 AC circuit containing only a capacitor

Consider a circuit containing a capacitor of capacitance C connected across an alternating voltage source (Figure 4.44). The instantaneous value of the alternating voltage is given by

$$v = V_m \sin \omega t \qquad (4.43)$$

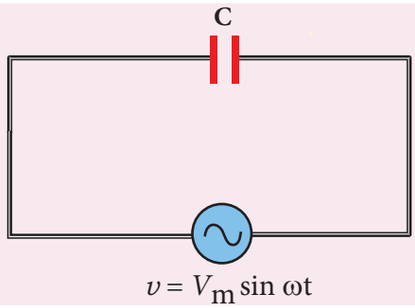


Figure 4.44 AC circuit with capacitor

Let q be the instantaneous charge on the capacitor. The emf across the capacitor at that instant is $\frac{q}{C}$. According to Kirchoff's loop rule,

$$v - \frac{q}{C} = 0$$

$$q = CV_m \sin \omega t$$

By the definition of current,

$$i = \frac{dq}{dt} = \frac{d}{dt}\left(CV_m \sin \omega t\right)$$

$$= CV_m \frac{d}{dt}\left(\sin \omega t\right)$$

$$= CV_m \omega \cos \omega t \qquad \text{(or)}$$

$$i = \frac{V_m}{1/C\omega} \sin\left(\omega t + \pi/2\right)$$

Instantaneous value of current,

$$i = I_m \sin\left(\omega t + \pi/2\right) \qquad (4.44)$$

where $\frac{V_m}{1/C\omega} = I_m$, the peak value of the alternating current. From equations (4.43) and (4.44), it is clear that current leads the applied voltage by $\pi/2$ in a capacitive circuit. This is shown pictorially in Figure 4.45. The wave diagram for a capacitive circuit also shows that the current leads the applied voltage by 90°.

Capacitive reactance X_C

The peak value of current I_m is given by $I_m = \frac{V_m}{1/C\omega}$. Let us compare this equation with $I_m = \frac{V_m}{R}$ for a resistive circuit. The quantity $1/C\omega$ plays the same role as the resistance R in resistive circuit. This is the resistance offered by the capacitor, called capacitive reactance (X_C). It measured in ohm.

$$X_c = \frac{1}{\omega C} \qquad (4.45)$$

The capacitive reactance (X_C) varies inversely as the frequency. For a steady current, $f = 0$.

$$\therefore X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$$

Thus a capacitive circuit offers infinite resistance to the steady current. So that steady current cannot flow through the capacitor.

12th Physics_Vol-1_Unit 4_English Medium.indd 241 23-12-2022 17:58:22